Motivation and Research Question

Question: How does accounting for imperfect competition in PF estimation change measured procurement markup premia?

Background

- Standard estimators (OP/LP/ACF) assume perfect competition (De Loecker & Syverson, 2021)
- ADL (2024): under oligopoly, input demand depends on competitive environment $\Rightarrow$ contaminates markups; bias is *differential* across treated/control $\Rightarrow$ distorts causal estimates
- **Base:** Chadim (2024, MSc thesis): DLW/ACF markups in Czech construction; procurement ATT $\approx 15\%$, declining $30\% \rightarrow 10\%$
- **This project:** ADL sufficient-statistic correction $\Rightarrow$ test robustness of ATT

Data: Czech construction (CZ-NACE F), 2005–2021

- **Financials** (MagnusWeb): 1,297 firms, 7,261 obs. | **Procurement** (Datlab): tenders, bidder identities

Approach and Contributions

Methodology

1. **Starting point** (Chadim, 2024)**:** ACF translog, perfect competition; $\hat{\mu}_{jt} = \hat{\theta}^V_{jt} / \alpha^V_{jt}$
2. **Imperfect competition** (ADL)**:** Condition control function on Q_{-jt} (Cournot) or Φ_{-jt} (logit); oligopoly instruments from $f_{-j,t}$ and $\omega_{-j,t-1}$
3. **Revenue validation:** QEC (ADL, §6.2); De Ridder et al. (2026)

Contributions

- First imperfect-competition PF estimation in procurement, with directly observed market structure
- Quantifies bias from perfect-competition assumption (Bond et al., 2021)
- Procurement markets provide natural competitive variation aiding PF identification
- Policy evaluation: transparency reforms (2012 single-bidding ban, 2017 Register of Contracts) as exogenous shifters

Referee reports: ADL (2024) [methodology] | Gibbon & Schain (2023) [empirical application]